

Bharath Kumar Kondur

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Summary

AI/ML Engineer with experience building production AI systems, GenAI applications, LLM-powered workflows, and scalable MLOps infrastructure. Strong background in real-time computer vision, edge AI deployment, TensorRT optimisation, async streaming systems, and cloud-native ML services using Docker, Kubernetes, FastAPI, MQTT, AWS, and Azure. Hands-on experience working with Claude, Codex, Cursor, LangChain, Hugging Face, and LLM-assisted development workflows for automation and AI engineering.

Experience

Project Consulting Srl | Rome, Italy

ML & Computer Vision Engineer | 11/2022 - Present

- Designed and deployed production AI pipelines for pose estimation, OCR, object detection, and streaming analytics.
- Optimised edge AI inference using ONNX, TensorRT, FP16/INT8 quantisation, and async processing pipelines.
- Built MLOps infrastructure using Docker, Kubernetes, FastAPI, MQTT, GitHub Actions, AWS, and Azure.
- Implemented Grafana and Prometheus observability systems for inference monitoring and anomaly detection.
- Developed scalable architectures for RTSP video analytics and IoT sensor stream processing.

Selected Projects

Scene Intelligence Platform: Multimodal perception platform combining object detection, tracking, OCR, pose estimation, and async event pipelines integrated with LLM.

Human Fall Detection: Production-ready fall detection system using YOLO pose models, temporal heuristics, and TensorRT optimisation.

Focus Tracker: Local-first productivity application with on-device ML intent classification and activity tracking.

AI Automation Tools: Built AI-integrated automation tools, developer utilities, and open-source projects published on GitHub.

TECHNICAL SKILLS

ML & AI: PyTorch, TensorFlow, Scikit-learn, Hugging Face, LoRA fine-tuning, LLMs, GenAI, LangChain, Claude, Codex, Cursor, prompt engineering, AI-assisted development

Computer Vision: YOLO11-pose, YOLO26l-pose, YOLOv8, OpenCV, OCR, ByteTrack, DeepSORT

MLOps & Infrastructure: Docker, Kubernetes, GitHub Actions, GitLab CI/CD, FastAPI, Linux, AWS

Edge AI & Optimisation: TensorRT, ONNX, FP16/INT8 quantisation, edge deployment, real-time inference

Data & Streaming: PostgreSQL, MongoDB, MinIO, MQTT, async Python, SQLAlchemy, WebSocket, REST APIs

Observability: Grafana, structured logging, alerting, inference monitoring

Education

Sapienza University of Rome | Rome, Italy

Master of Science in Energy Engineering | 07/2022

Thesis: Improving Accuracy in LCOE Estimation using Machine Learning

JNTUH, Hyderabad | India

Bachelor of Technology in Electrical & Electronics Engineering | 06/2016

Languages

English, Italian, Hindi, Telugu